

Exercise 4

Calculation of the water surface profile upstream of a dam

One purpose for which dams are designed and built is flood protection. Among the range of criteria that need to be considered, the design engineer must ascertain that the dam height is sufficient for the annual exceedance probability of the design flood, and the design engineer must also ascertain that the change in water level upstream of the dam does not result in negative consequences. You will investigate these two aspects in detail. For the channel under consideration with trapezoidal geometry (base width $b = 10$ m, side slope $1:m = 1:2$ and channel bed slope $S_0 = 0.00025$), compute the water surface profile for a design discharge $Q = 25$ m³/s and a Manning's roughness coefficient $n = 0.02857$ s/m^{1/3}. The flow is subcritical with a uniform flow depth $y_0 = 2.25$ m. The proposed dam blocks the water at $x = 0$ m up to an elevation $y_p = 382$ m above mean sea level with a water depth at the dam of $y = 3.10$ m.

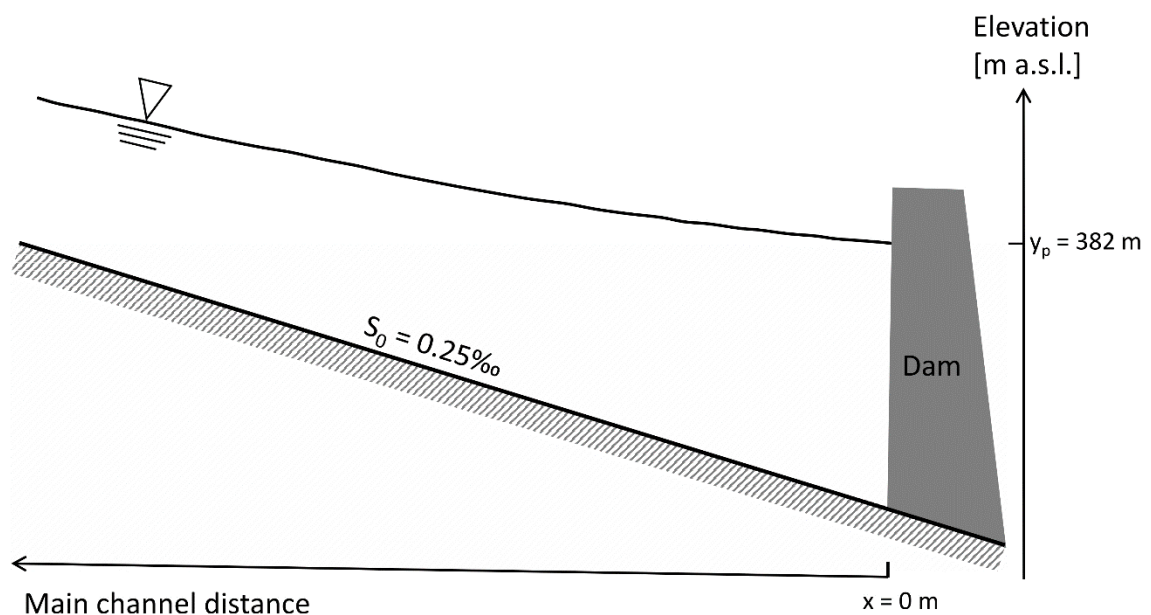


Figure 1a: Longitudinal-section view of the trapezoidal channel

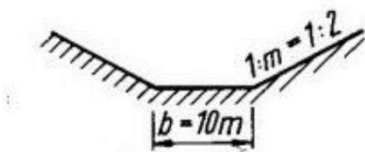


Figure 1b: Cross-section view of the trapezoidal channel

Determine the water surface profile using the standard step method.

Solution

Calculations for first cross section:

Governing equation:

$$\Delta y_p = \Delta y - S_0 \Delta x = -\frac{\bar{u}^2 n^2}{\bar{R}^{4/3}} \Delta x - \frac{\bar{u}}{g} \cdot \Delta u$$

- Δy_p : change in piezometric (hydraulic) head relative to datum, e.g. a.s.l.
- Δx : length of channel section over which Δy_p occurs.
- u : flow velocity; $\bar{u}$: average velocity for section; Δu : velocity difference
- $\bar{R}$: average hydraulic radius for cross section Δx
- Note: Expansion or contraction losses are not considered here.

Solution:

Remark:

- This example could also be calculated using the direct step method, which does not require iteration. The direct step method is applicable here since the cross-section geometry is the same for each cross-section (prismatic channel). However, in order to gain insight into how the standard step method is applied, an example with prismatic geometry was selected.
- Water surface profile type: M1 ($y_0 > y_c$, backwater)
- Starting point for calculation, direction? $\rightarrow$ conditions known directly upstream of the dam, so that is where the starting point is. Calculations carried out from downstream to upstream (subcritical conditions).
- Required set of equations:

$$\text{Area: } A = y \cdot (b + m \cdot y) = y \cdot (10 + 2 \cdot y)$$

$$\text{Wetted perimeter: } P = b + 2 \cdot y \cdot \sqrt{1 + m^2} = 10 + 2 \cdot \sqrt{5} \cdot y$$

$$\text{Hydraulic radius: } R = \frac{A}{P}$$

$$\text{Flow velocity: } u = \frac{Q}{A} = \frac{25}{y(10+2y)}$$

Change in piezometric head (with respect to reference height, i.e. datum, e.g. a.s.l.) over the section Δx :

$$\Delta y_p = \frac{u_1^2}{2g} - \frac{u_2^2}{2g} - h_L$$

$$S_f = \frac{Q^2 n^2}{A^2 R^{4/3}} = \frac{u^2 n^2}{R^{4/3}}$$

$$\text{and } \Delta y_p = \frac{u_1^2}{2g} - \frac{u_2^2}{2g} - \bar{S}_f \Delta x$$

$$\Delta y = \Delta y_p + S_0 \Delta x$$

$$\Delta y = y_i - y_{i-1}$$

Working equation for Δy_p :

$$\Delta y_p = -\frac{\bar{u}^2 n^2}{\bar{R}^{4/3}} \Delta x - \frac{\frac{u_1 + u_2}{2}}{g} (u_2 - u_1)$$

$$\text{with: } \bar{u} = \frac{u_1 + u_2}{2}; \Delta u = u_2 - u_1; \bar{R} = \frac{R_1 + R_2}{2}$$

Start at the dam (x = 0 m):

$$A_i = 3.1(10 + 2 \times 3.1) = 50.22 \text{ m}^2$$

$$P_i = 10 + \sqrt{5} \times 3.1 = 23.86 \text{ m}$$

$$R_i = \frac{50.22}{23.86} = 2.104 \text{ m}$$

$$u_i = \frac{Q}{A} = \frac{25}{50.22} = 0.498 \text{ m/s}$$

Calculation of the water depth y_1 at 1225 m upstream:

1st guess: $\Delta y = 0.1 \text{ m}$

At cross section 1:

$$A_{i-1} = 3.0(10 + 2 \times 3.0) = 48.00 \text{ m}^2$$

$$P_{i-1} = 10 + \sqrt{5} \times 3.0 = 23.42 \text{ m}$$

$$R_{i-1} = \frac{48.00}{23.42} = 2.05 \text{ m}$$

$$u_{i-1} = \frac{Q}{A} = \frac{25}{48.00} = 0.521 \text{ m/s}$$

Average values between cross sections 0 and 1:

$$\bar{R} = \frac{2.05 + 2.10}{2} = 2.075 \text{ m}$$

$$\bar{u} = \frac{0.521 + 0.498}{2} = 0.5095 \text{ m/s}$$

$$\text{Velocity difference: } \Delta u = u_i - u_{i-1} = 0.498 - 0.521 = -0.023 \text{ m/s}$$

Change in piezometric head:

$$\Delta y_p = -\frac{0.5095^2 \times 0.02857^2}{2.075^{4/3}} \times 1225 - \frac{0.5095}{9.81}(-0.023) = -0.097 \text{ m}$$

Change in water depth:

$$\Delta y = -0.097 + 0.00025 \times 1225 = 0.21 \text{ m} > 0.1 \text{ m} (= 1st \text{ guess})$$

A next iteration required, using a larger Δy .

2nd guess: $\Delta y = 0.204 \text{ m}$

At cross section 1:

$$h_{i-1} = 3.1 - 0.204 = 2.896 \text{ m}$$

$$A_{i-1} = 2.896(10 + 2 \times 2.896) = 45.73 \text{ m}^2$$

$$P_{i-1} = 10 + \sqrt{5} \times 2.896 = 22.95 \text{ m}$$

$$R_{i-1} = \frac{45.73}{22.95} = 1.993 \text{ m}$$

$$u_{i-1} = \frac{Q}{A} = \frac{25}{45.73} = 0.547 \text{ m/s}$$

Average values between cross sections 0 and 1:

$$\bar{R} = \frac{1.993 + 2.104}{2} = 2.049 \text{ m}$$

$$\bar{u} = \frac{0.547 + 0.498}{2} = 0.522 \text{ m/s}$$

$$\text{Velocity difference: } \Delta u = u_i - u_{i-1} = 0.498 - 0.547 = -0.049 \text{ m/s}$$

Change in piezometric head:

$$\Delta y_p = -\frac{0.5095^2 \times 0.02857^2}{2.049^{4/3}} \times 1225 - \frac{0.522}{9.81}(-0.049) = -0.097 \text{ m}$$

Change in water depth:

$$\Delta y = -0.097 + 0.00025 \times 1225 = 0.21 \text{ m} \approx 0.204 \text{ m} (= 2nd \text{ guess})$$

Solution is close to the guessed value. Continue with next calculations for next cross sections (not shown here).